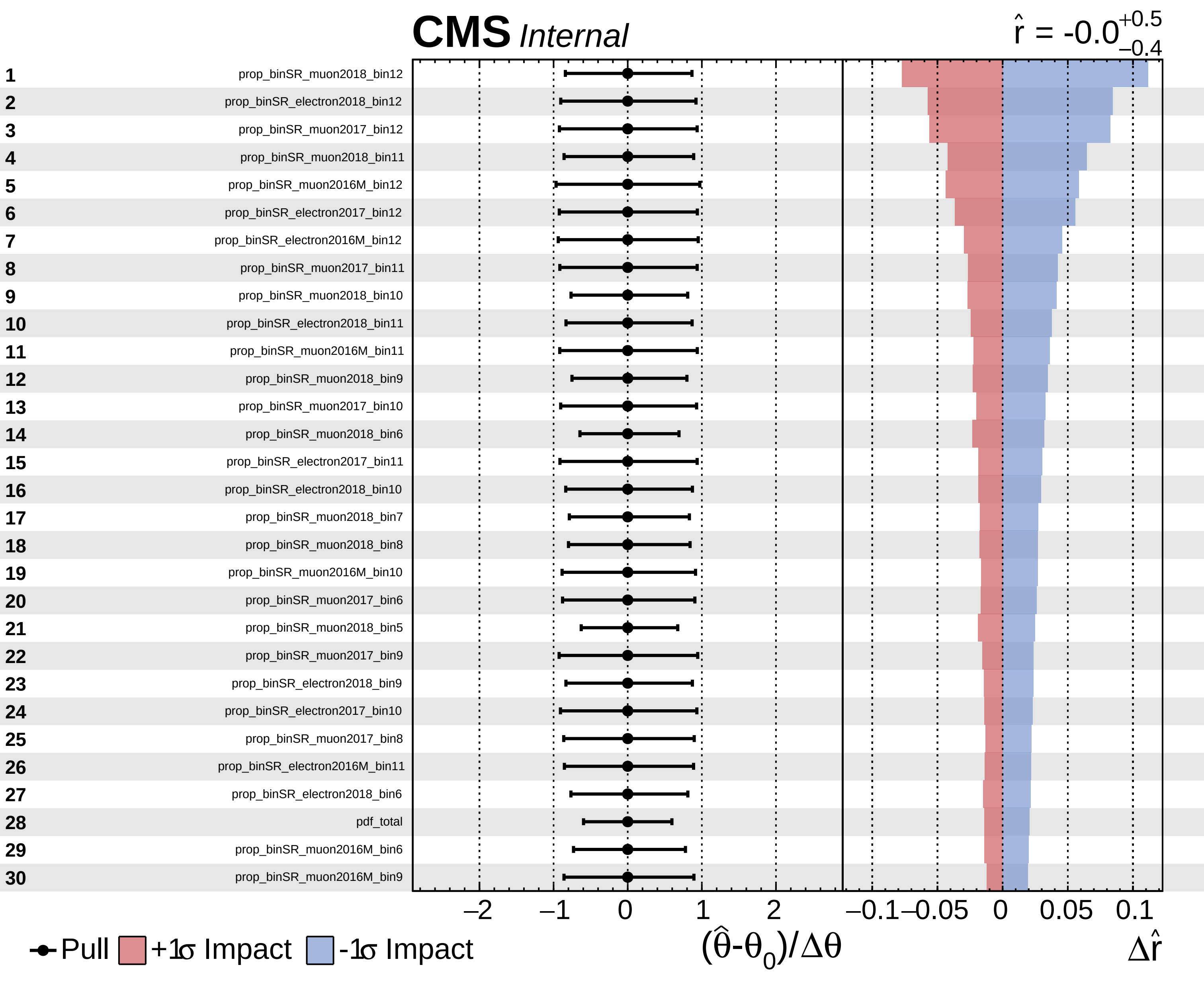
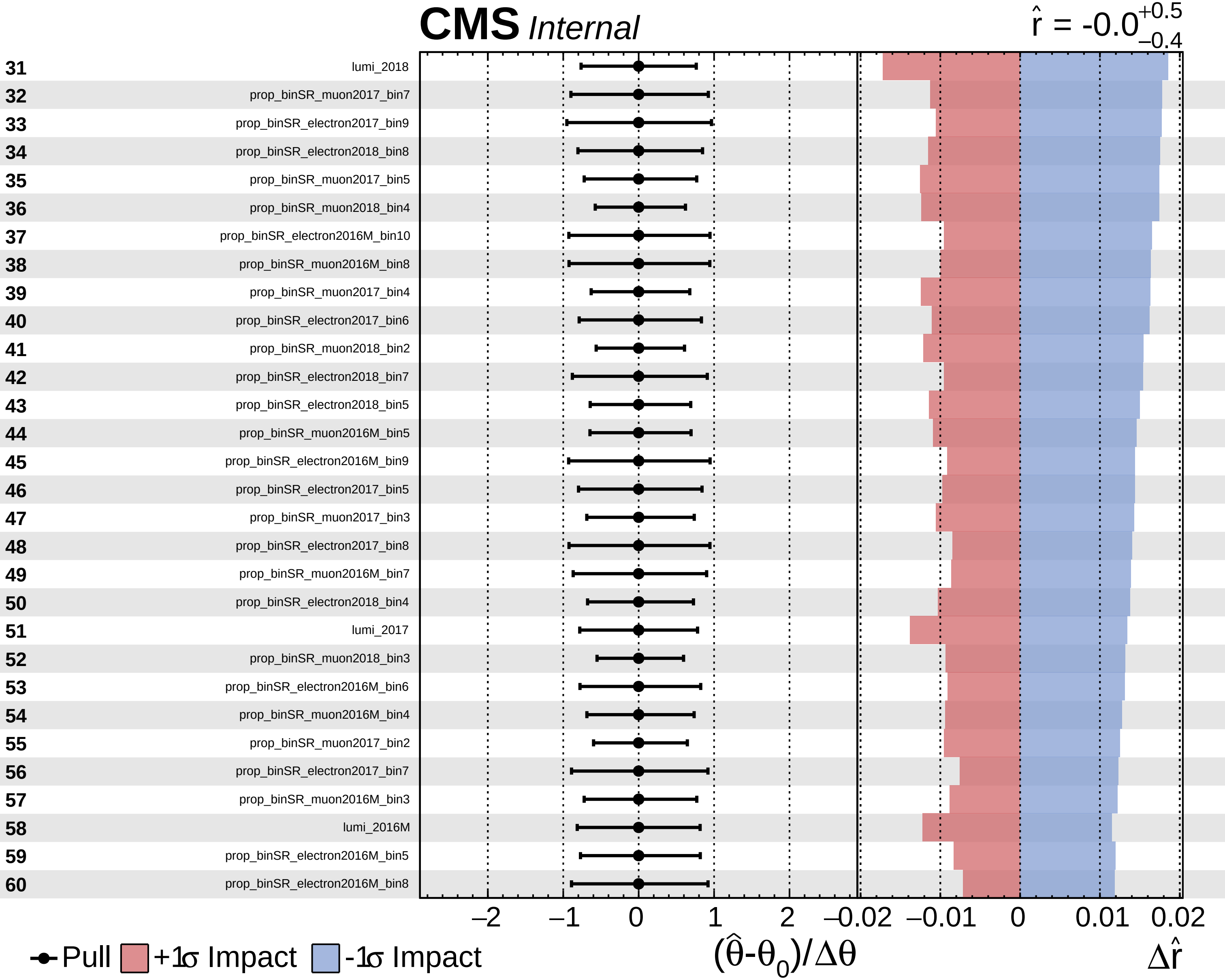
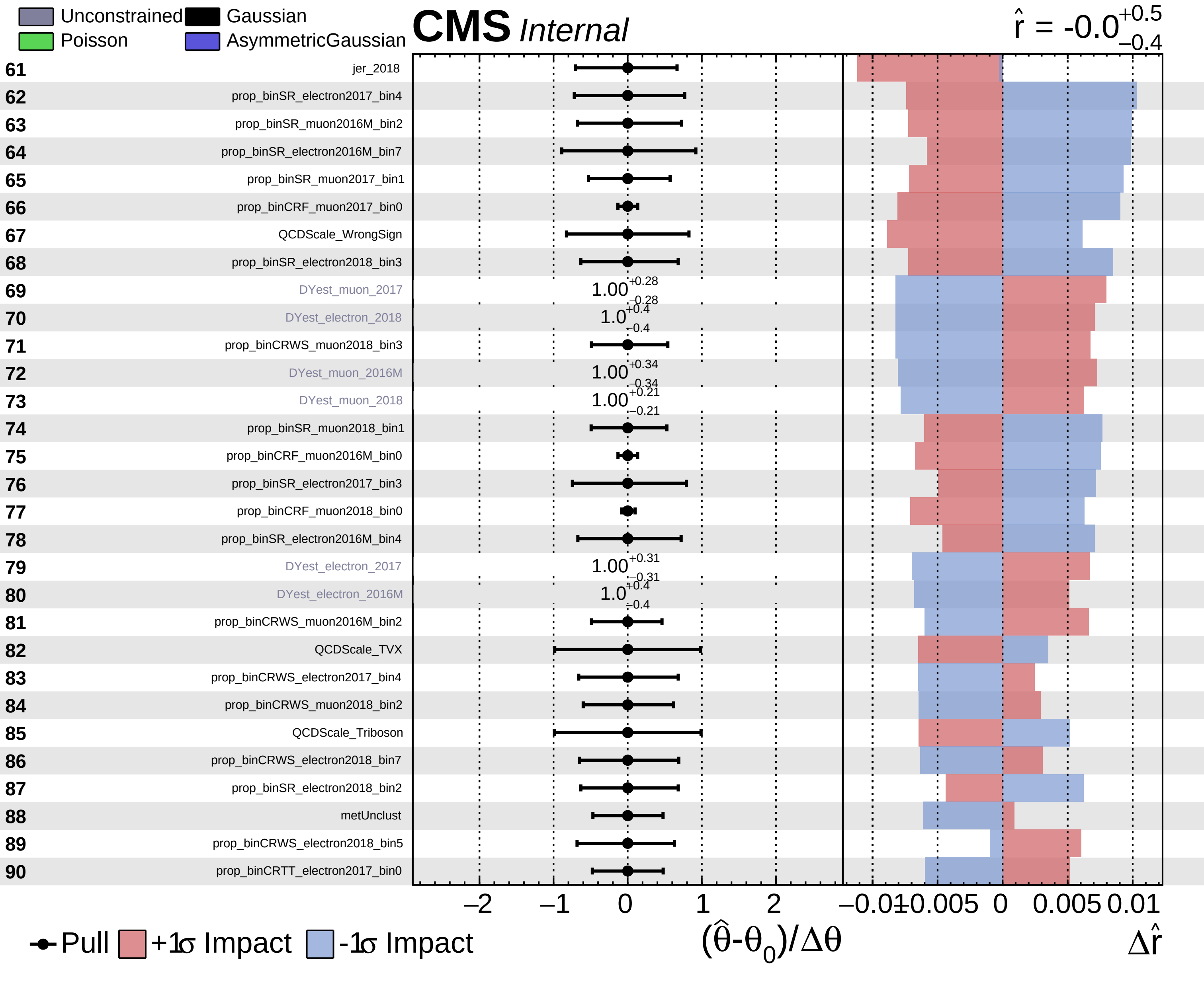


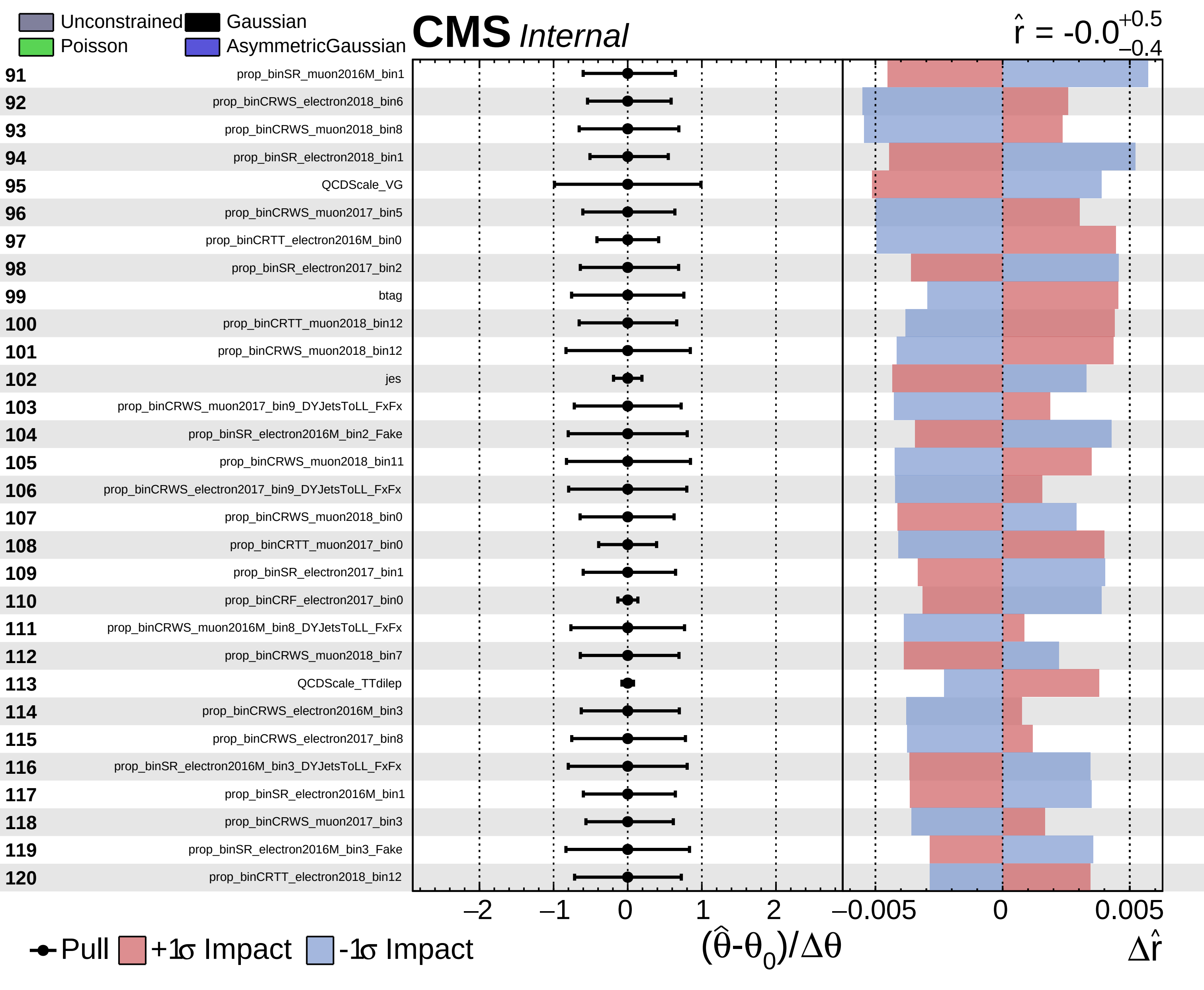
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
1 prop_binSR_muon2018_bin12
2 prop_binSR_electron2018_bin12
3 prop_binSR_muon2017_bin12
4 prop_binSR_muon2018_bin11
5 prop_binSR_muon2016M_bin12
6 prop_binSR_electron2017_bin12
7 prop_binSR_electron2016M_bin12
8 prop_binSR_muon2017_bin11
9 prop_binSR_muon2018_bin10
10 prop_binSR_electron2018_bin11
11 prop_binSR_muon2016M_bin11
12 prop_binSR_muon2018_bin9
13 prop_binSR_muon2017_bin10
14 prop_binSR_muon2018_bin6
15 prop_binSR_electron2017_bin11
16 prop_binSR_electron2018_bin10
17 prop_binSR_muon2018_bin7
18 prop_binSR_muon2018_bin8
19 prop_binSR_muon2016M_bin10
20 prop_binSR_muon2017_bin6
21 prop_binSR_muon2018_bin5
22 prop_binSR_muon2017_bin9
23 prop_binSR_electron2018_bin9
24 prop_binSR_electron2017_bin10
25 prop_binSR_muon2017_bin8
26 prop_binSR_electron2016M_bin11
27 prop_binSR_electron2018_bin6
28 pdf_total
29 prop_binSR_muon2016M_bin6
30 prop_binSR_muon2016M_bin9
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

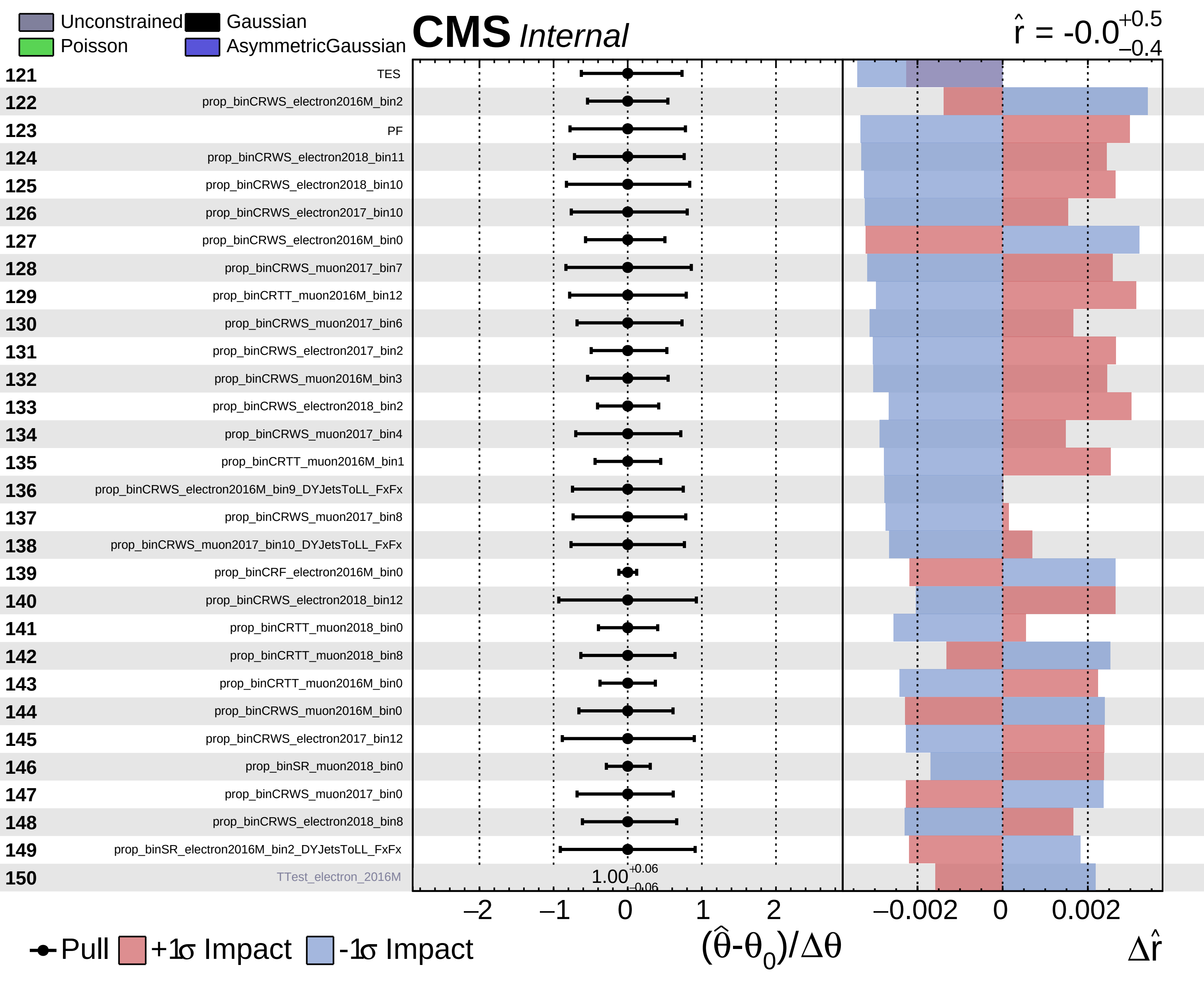


CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
31 lumi_2018
32 prop_binSR_muon2017_bin7
33 prop_binSR_electron2017_bin9
34 prop_binSR_electron2018_bin8
35 prop_binSR_muon2017_bin5
36 prop_binSR_muon2018_bin4
37 prop_binSR_electron2016M_bin10
38 prop_binSR_muon2016M_bin8
39 prop_binSR_muon2017_bin4
40 prop_binSR_electron2017_bin6
41 prop_binSR_muon2018_bin2
42 prop_binSR_electron2018_bin7
43 prop_binSR_electron2018_bin5
44 prop_binSR_muon2016M_bin5
45 prop_binSR_electron2016M_bin9
46 prop_binSR_electron2017_bin5
47 prop_binSR_muon2017_bin3
48 prop_binSR_electron2017_bin8
49 prop_binSR_muon2016M_bin7
50 prop_binSR_electron2018_bin4
51 lumi_2017
52 prop_binSR_muon2018_bin3
53 prop_binSR_electron2016M_bin6
54 prop_binSR_muon2016M_bin4
55 prop_binSR_muon2017_bin2
56 prop_binSR_electron2017_bin7
57 prop_binSR_muon2016M_bin3
58 lumi_2016M
59 prop_binSR_electron2016M_bin5
60 prop_binSR_electron2016M_bin8
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

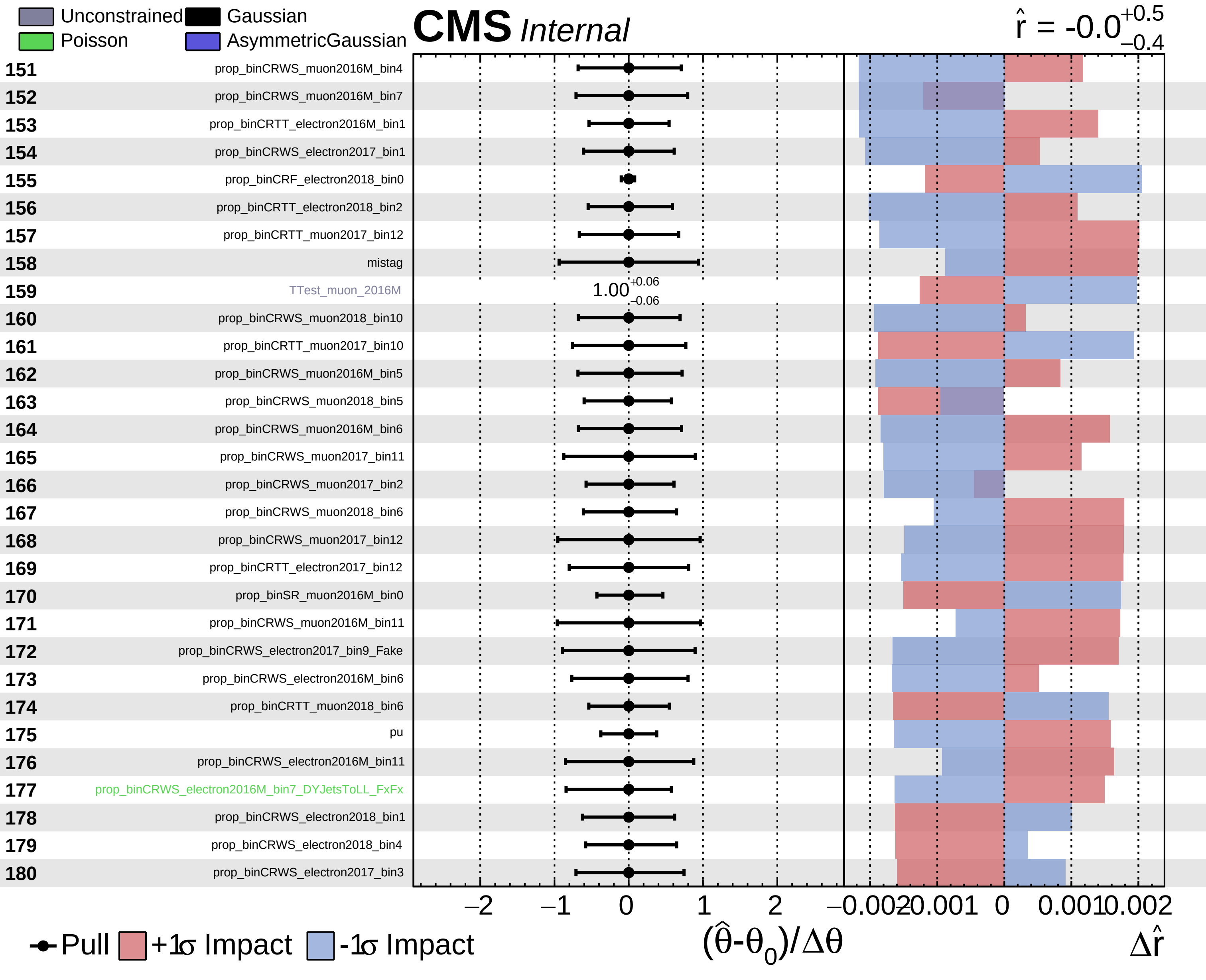


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
61 jer_2018
62 prop_binSR_electron2017_bin4
63 prop_binSR_muon2016M_bin2
64 prop_binSR_electron2016M_bin7
65 prop_binSR_muon2017_bin1
66 prop_binCRF_muon2017_bin0
67 QCDScale_WrongSign
68 prop_binSR_electron2018_bin3
69 DYest_muon_2017 $1.00^{+0.28}_{-0.28}$
70 DYest_electron_2018 $1.0^{+0.4}_{-0.4}$
71 prop_binCRWS_muon2018_bin3
72 DYest_muon_2016M $1.00^{+0.34}_{-0.34}$
73 DYest_muon_2018 $1.00^{+0.21}_{-0.21}$
74 prop_binSR_muon2018_bin1
75 prop_binCRF_muon2016M_bin0
76 prop_binSR_electron2017_bin3
77 prop_binCRF_muon2018_bin0
78 prop_binSR_electron2016M_bin4
79 DYest_electron_2017 $1.00^{+0.31}_{-0.31}$
80 DYest_electron_2016M $1.0^{+0.4}_{-0.4}$
81 prop_binCRWS_muon2016M_bin2
82 QCDScale_TVX
83 prop_binCRWS_electron2017_bin4
84 prop_binCRWS_muon2018_bin2
85 QCDScale_Triboson
86 prop_binCRWS_electron2018_bin7
87 prop_binSR_electron2018_bin2
88 metUnclust
89 prop_binCRWS_electron2018_bin5
90 prop_binCRTT_electron2017_bin0
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

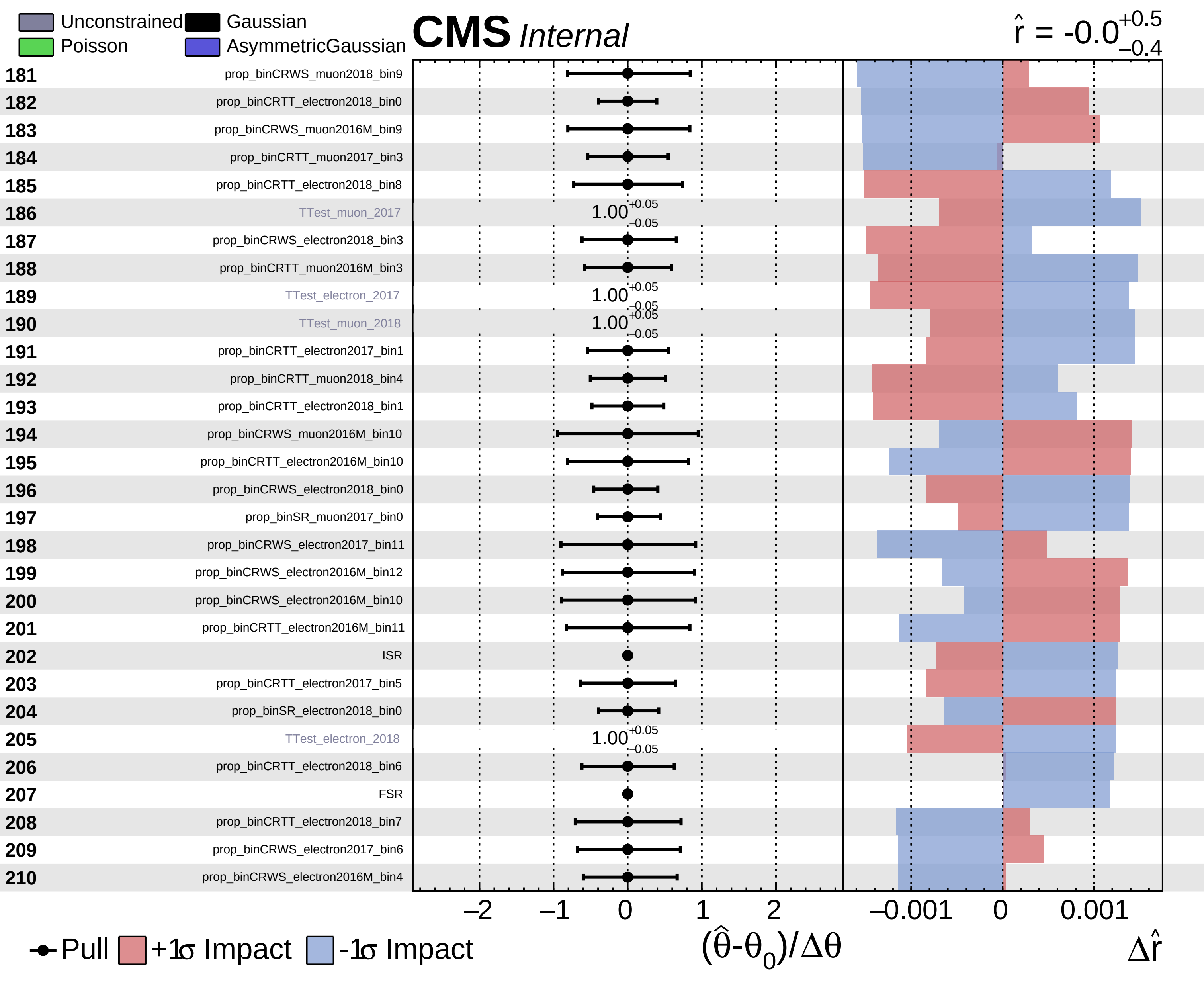

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
91 prop_binSR_muon2016M_bin1
92 prop_binCRWS_electron2018_bin6
93 prop_binCRWS_muon2018_bin8
94 prop_binSR_electron2018_bin1
95 QCDScale_VG
96 prop_binCRWS_muon2017_bin5
97 prop_binCRTT_electron2016M_bin0
98 prop_binSR_electron2017_bin2
99 btag
100 prop_binCRTT_muon2018_bin12
101 prop_binCRWS_muon2018_bin12
102 jes
103 prop_binCRWS_muon2017_bin9_DYJetsToLL_FxFx
104 prop_binSR_electron2016M_bin2_Fake
105 prop_binCRWS_muon2018_bin11
106 prop_binCRWS_electron2017_bin9_DYJetsToLL_FxFx
107 prop_binCRWS_muon2018_bin0
108 prop_binCRTT_muon2017_bin0
109 prop_binSR_electron2017_bin1
110 prop_binCRF_electron2017_bin0
111 prop_binCRWS_muon2016M_bin8_DYJetsToLL_FxFx
112 prop_binCRWS_muon2018_bin7
113 QCDScale_TTdilep
114 prop_binCRWS_electron2016M_bin3
115 prop_binCRWS_electron2017_bin8
116 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
117 prop_binSR_electron2016M_bin1
118 prop_binCRWS_muon2017_bin3
119 prop_binSR_electron2016M_bin3_Fake
120 prop_binCRTT_electron2018_bin12
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



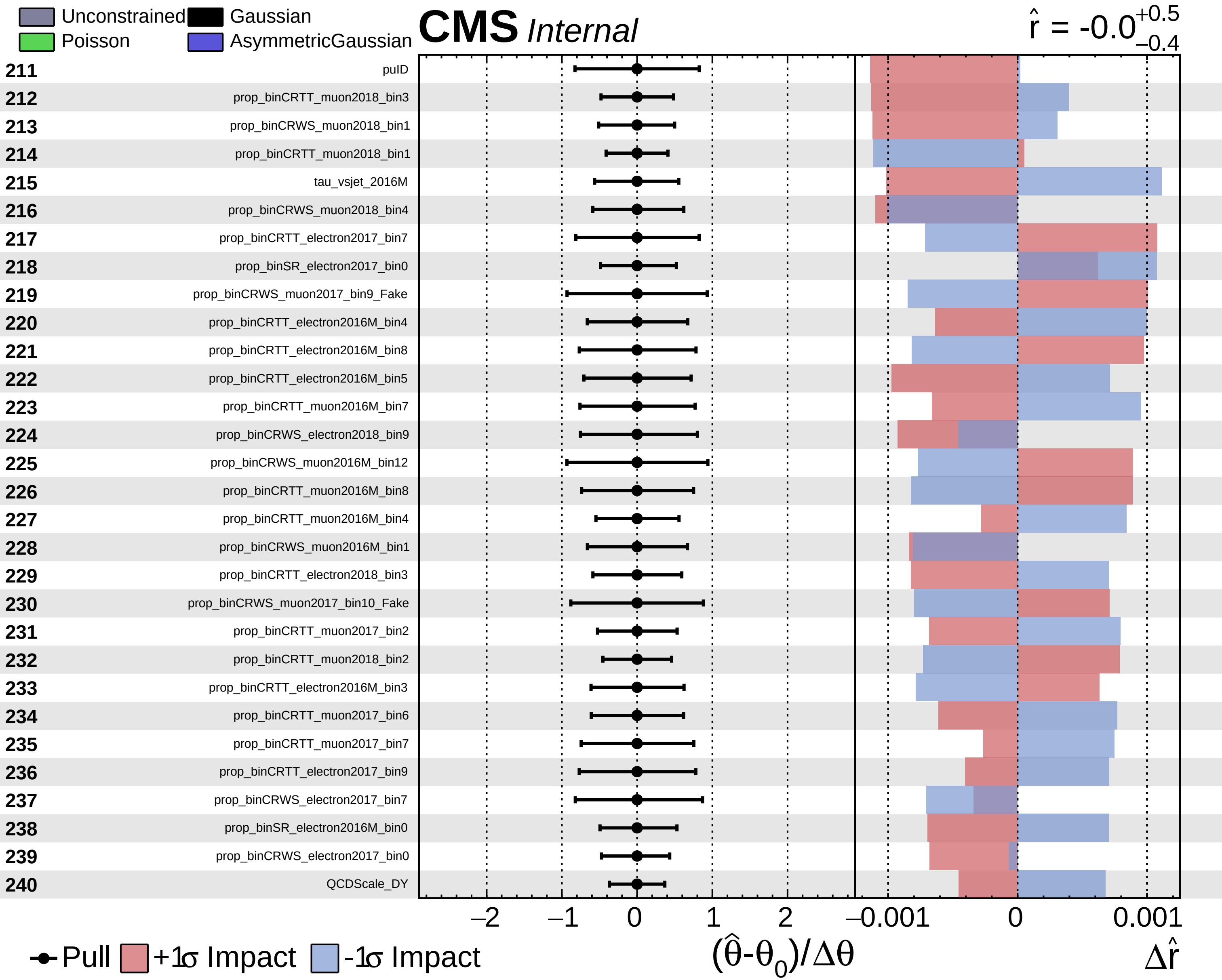
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
121 TES
122 prop_binCRWS_electron2016M_bin2
123 PF
124 prop_binCRWS_electron2018_bin11
125 prop_binCRWS_electron2018_bin10
126 prop_binCRWS_electron2017_bin10
127 prop_binCRWS_electron2016M_bin0
128 prop_binCRWS_muon2017_bin7
129 prop_binCRTT_muon2016M_bin12
130 prop_binCRWS_muon2017_bin6
131 prop_binCRWS_electron2017_bin2
132 prop_binCRWS_muon2016M_bin3
133 prop_binCRWS_electron2018_bin2
134 prop_binCRWS_muon2017_bin4
135 prop_binCRTT_muon2016M_bin1
136 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
137 prop_binCRWS_muon2017_bin8
138 prop_binCRWS_muon2017_bin10_DYJetsToLL_FxFx
139 prop_binCRF_electron2016M_bin0
140 prop_binCRWS_electron2018_bin12
141 prop_binCRTT_muon2018_bin0
142 prop_binCRTT_muon2018_bin8
143 prop_binCRTT_muon2016M_bin0
144 prop_binCRWS_muon2016M_bin0
145 prop_binCRWS_electron2017_bin12
146 prop_binSR_muon2018_bin0
147 prop_binCRWS_muon2017_bin0
148 prop_binCRWS_electron2018_bin8
149 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
150 TTest_electron_2016M
$1.00^{+0.06}_{-0.06}$
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

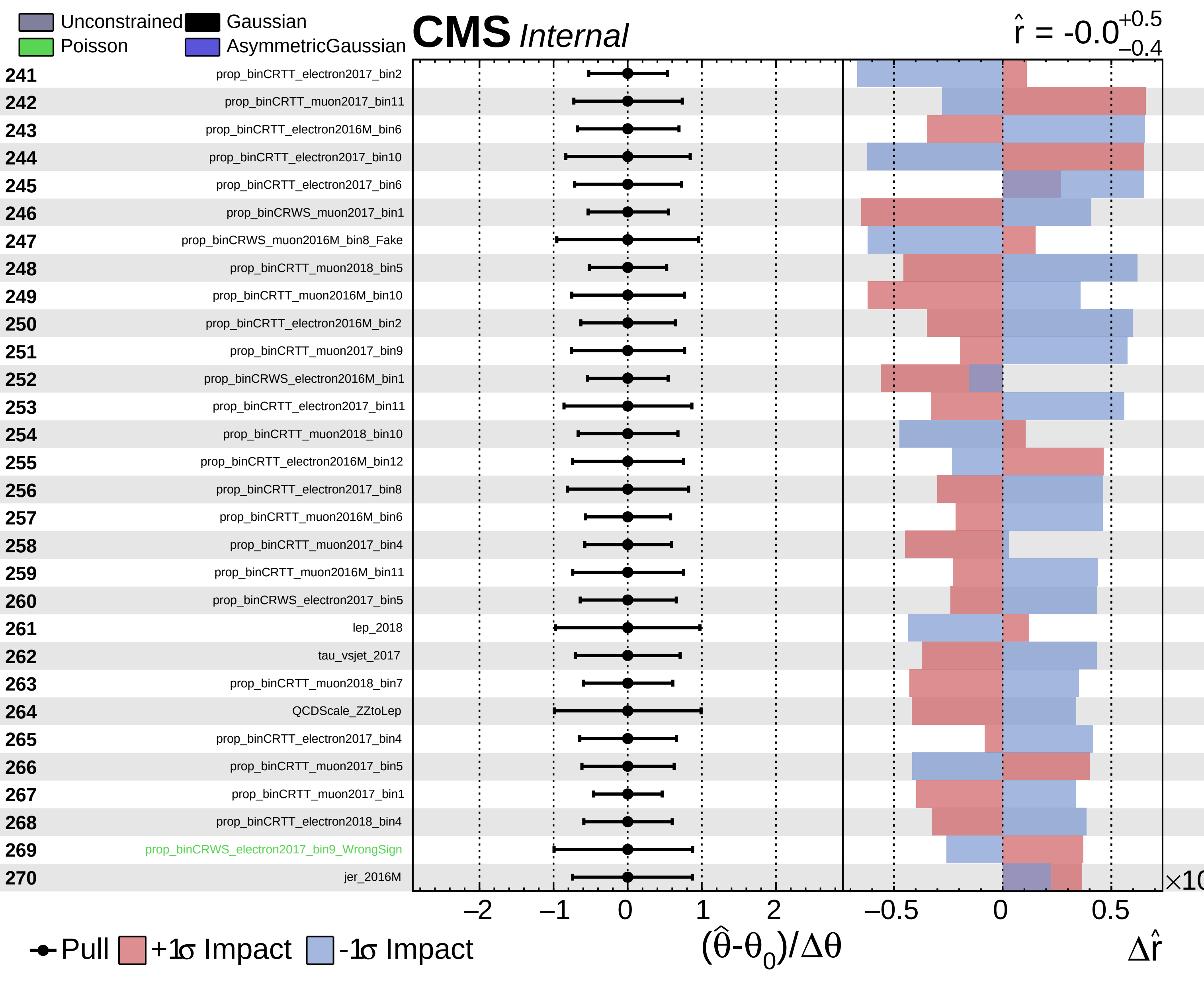


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
151 prop_binCRWS_muon2016M_bin4
152 prop_binCRWS_muon2016M_bin7
153 prop_binCRTT_electron2016M_bin1
154 prop_binCRWS_electron2017_bin1
155 prop_binCRF_electron2018_bin0
156 prop_binCRTT_electron2018_bin2
157 prop_binCRTT_muon2017_bin12
158 mistag
159 TTest_muon_2016M
$1.00^{+0.06}_{-0.06}$
160 prop_binCRWS_muon2018_bin10
161 prop_binCRTT_muon2017_bin10
162 prop_binCRWS_muon2016M_bin5
163 prop_binCRWS_muon2018_bin5
164 prop_binCRWS_muon2016M_bin6
165 prop_binCRWS_muon2017_bin11
166 prop_binCRWS_muon2017_bin2
167 prop_binCRWS_muon2018_bin6
168 prop_binCRWS_muon2017_bin12
169 prop_binCRTT_electron2017_bin12
170 prop_binSR_muon2016M_bin0
171 prop_binCRWS_muon2016M_bin11
172 prop_binCRWS_electron2017_bin9_Fake
173 prop_binCRWS_electron2016M_bin6
174 prop_binCRTT_muon2018_bin6
175 pu
176 prop_binCRWS_electron2016M_bin11
177 prop_binCRWS_electron2016M_bin7_DYJetsToLL_FxFx
178 prop_binCRWS_electron2018_bin1
179 prop_binCRWS_electron2018_bin4
180 prop_binCRWS_electron2017_bin3
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

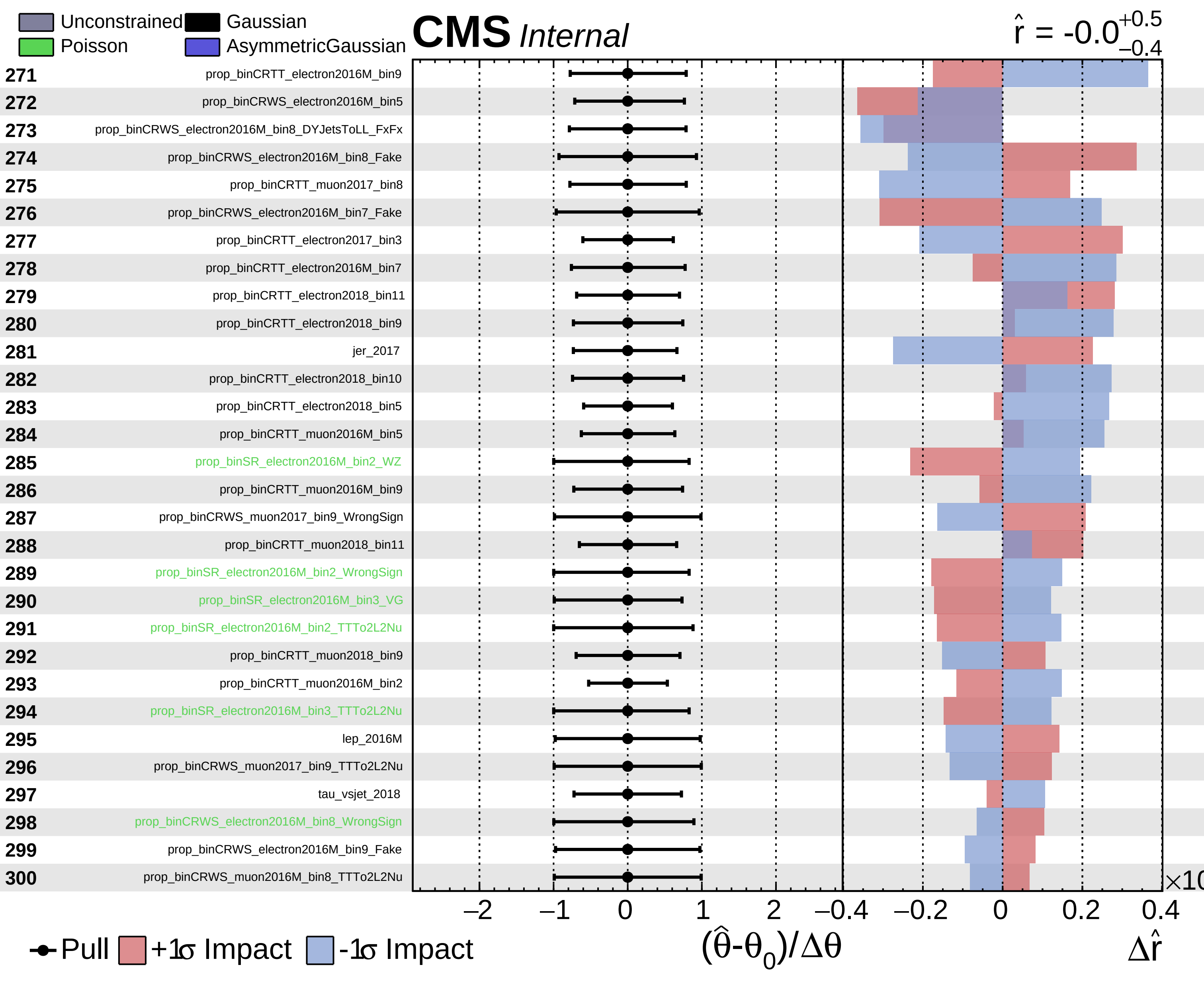


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.0+0.5 −0.4
181 prop_binCRWS_muon2018_bin9
182 prop_binCRTT_electron2018_bin0
183 prop_binCRWS_muon2016M_bin9
184 prop_binCRTT_muon2017_bin3
185 prop_binCRTT_electron2018_bin8
186 TTest_muon_2017
1.00+0.05 −0.05
187 prop_binCRWS_electron2018_bin3
188 prop_binCRTT_muon2016M_bin3
189 TTest_electron_2017
1.00+0.05 −0.05
190 TTest_muon_2018
1.00+0.05 −0.05
191 prop_binCRTT_electron2017_bin1
192 prop_binCRTT_muon2018_bin4
193 prop_binCRTT_electron2018_bin1
194 prop_binCRWS_muon2016M_bin10
195 prop_binCRTT_electron2016M_bin10
196 prop_binCRWS_electron2018_bin0
197 prop_binSR_muon2017_bin0
198 prop_binCRWS_electron2017_bin11
199 prop_binCRWS_electron2016M_bin12
200 prop_binCRWS_electron2016M_bin10
201 prop_binCRTT_electron2016M_bin11
202 ISR
203 prop_binCRTT_electron2017_bin5
204 prop_binSR_electron2018_bin0
205 TTest_electron_2018
1.00+0.05 −0.05
206 prop_binCRTT_electron2018_bin6
207 FSR
208 prop_binCRTT_electron2018_bin7
209 prop_binCRWS_electron2017_bin6
210 prop_binCRWS_electron2016M_bin4
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

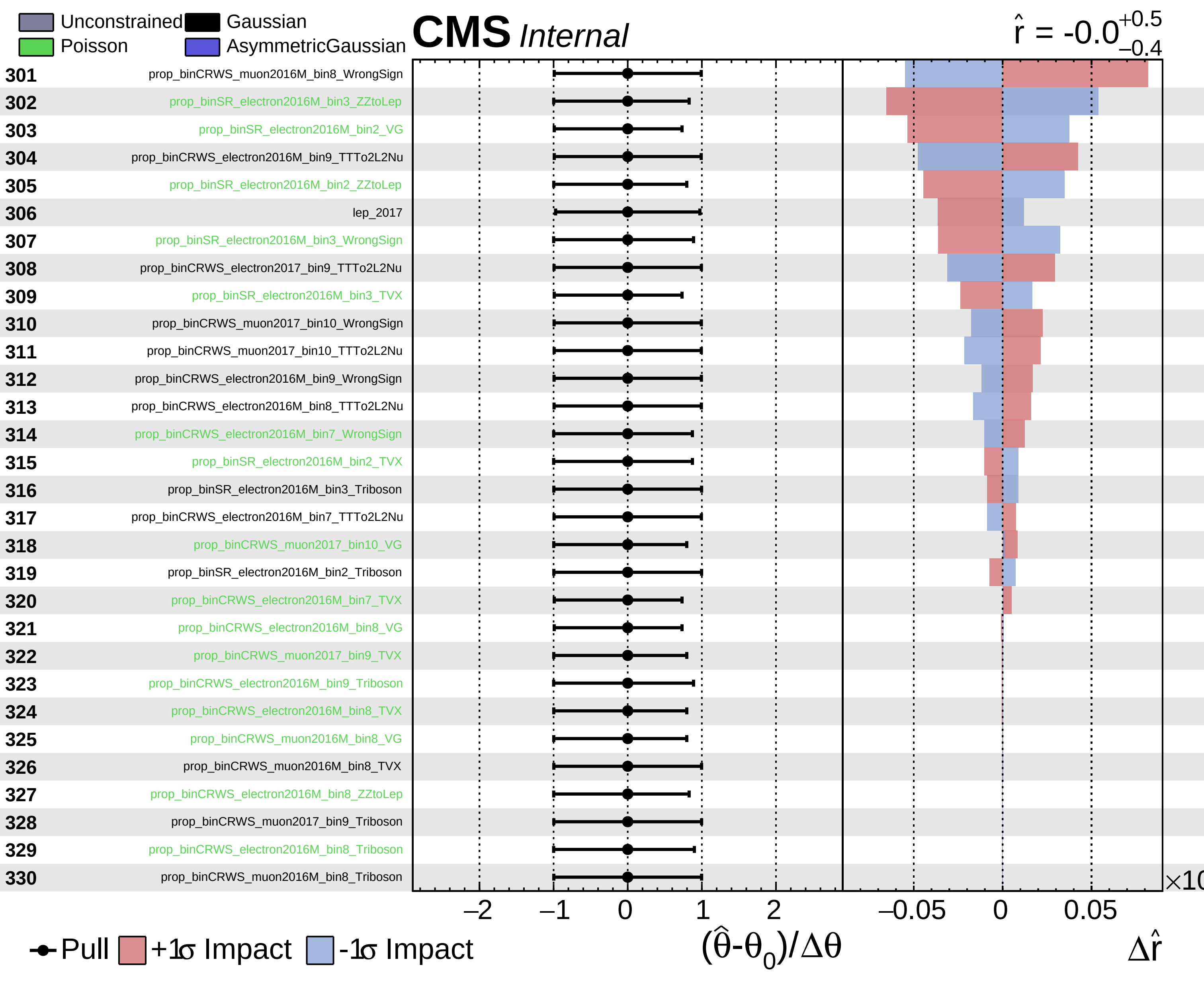

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
211 puID
212 prop_binCRTT_muon2018_bin3
213 prop_binCRWS_muon2018_bin1
214 prop_binCRTT_muon2018_bin1
215 tau_vsjet_2016M
216 prop_binCRWS_muon2018_bin4
217 prop_binCRTT_electron2017_bin7
218 prop_binSR_electron2017_bin0
219 prop_binCRWS_muon2017_bin9_Fake
220 prop_binCRTT_electron2016M_bin4
221 prop_binCRTT_electron2016M_bin8
222 prop_binCRTT_electron2016M_bin5
223 prop_binCRTT_muon2016M_bin7
224 prop_binCRWS_electron2018_bin9
225 prop_binCRWS_muon2016M_bin12
226 prop_binCRTT_muon2016M_bin8
227 prop_binCRTT_muon2016M_bin4
228 prop_binCRWS_muon2016M_bin1
229 prop_binCRTT_electron2018_bin3
230 prop_binCRWS_muon2017_bin10_Fake
231 prop_binCRTT_muon2017_bin2
232 prop_binCRTT_muon2018_bin2
233 prop_binCRTT_electron2016M_bin3
234 prop_binCRTT_muon2017_bin6
235 prop_binCRTT_muon2017_bin7
236 prop_binCRTT_electron2017_bin9
237 prop_binCRWS_electron2017_bin7
238 prop_binSR_electron2016M_bin0
239 prop_binCRWS_electron2017_bin0
240 QCDScale_DY
−2 −1 0 1 2
−0.001 0 0.001
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



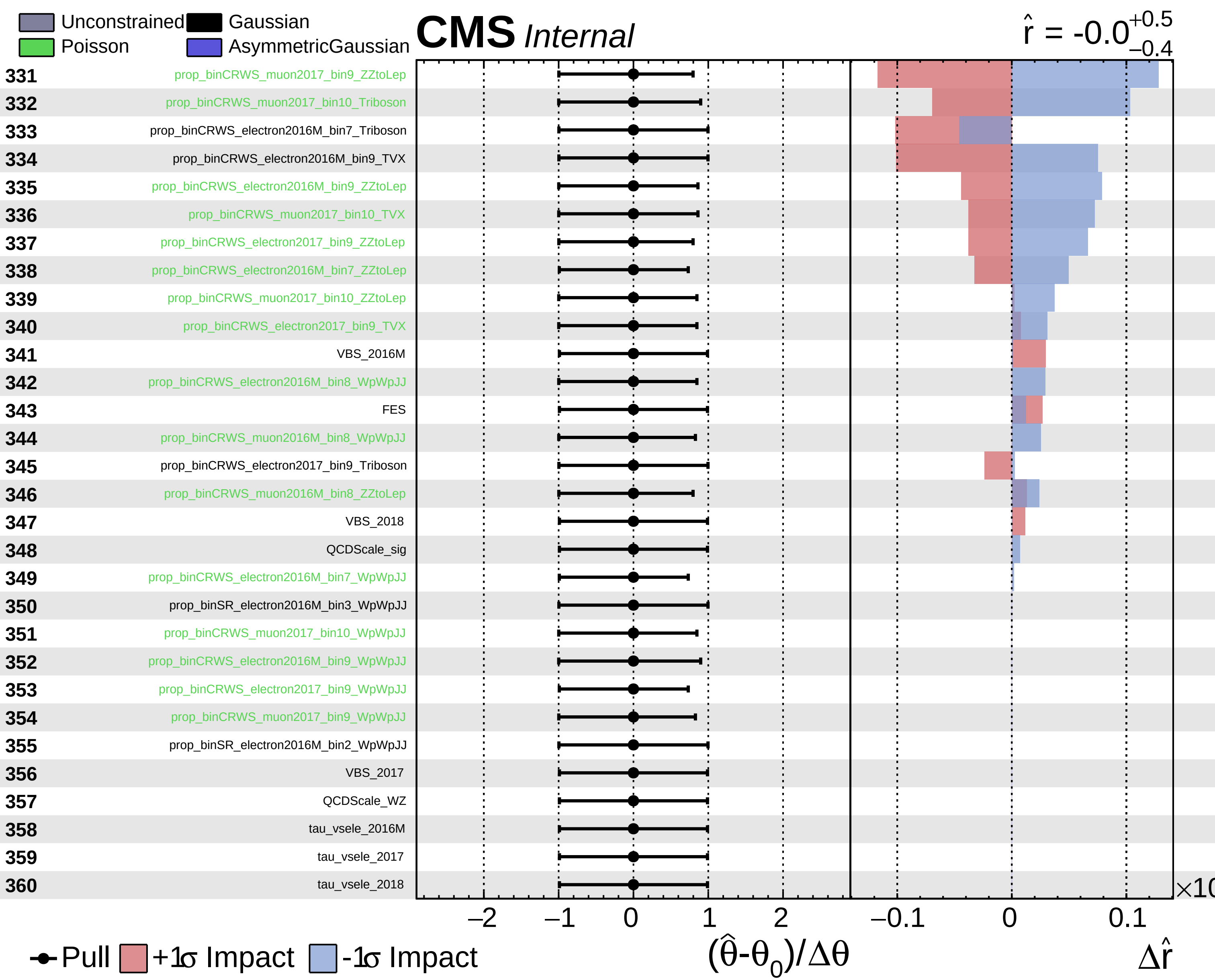
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
241 prop_binCRTT_electron2017_bin2
242 prop_binCRTT_muon2017_bin11
243 prop_binCRTT_electron2016M_bin6
244 prop_binCRTT_electron2017_bin10
245 prop_binCRTT_electron2017_bin6
246 prop_binCRWS_muon2017_bin1
247 prop_binCRWS_muon2016M_bin8_Fake
248 prop_binCRTT_muon2018_bin5
249 prop_binCRTT_muon2016M_bin10
250 prop_binCRTT_electron2016M_bin2
251 prop_binCRTT_muon2017_bin9
252 prop_binCRWS_electron2016M_bin1
253 prop_binCRTT_electron2017_bin11
254 prop_binCRTT_muon2018_bin10
255 prop_binCRTT_electron2016M_bin12
256 prop_binCRTT_electron2017_bin8
257 prop_binCRTT_muon2016M_bin6
258 prop_binCRTT_muon2017_bin4
259 prop_binCRTT_muon2016M_bin11
260 prop_binCRWS_electron2017_bin5
261 lep_2018
262 tau_vsjet_2017
263 prop_binCRTT_muon2018_bin7
264 QCDScale_ZZtoLep
265 prop_binCRTT_electron2017_bin4
266 prop_binCRTT_muon2017_bin5
267 prop_binCRTT_muon2017_bin1
268 prop_binCRTT_electron2018_bin4
269 prop_binCRWS_electron2017_bin9_WrongSign
270 jer_2016M
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



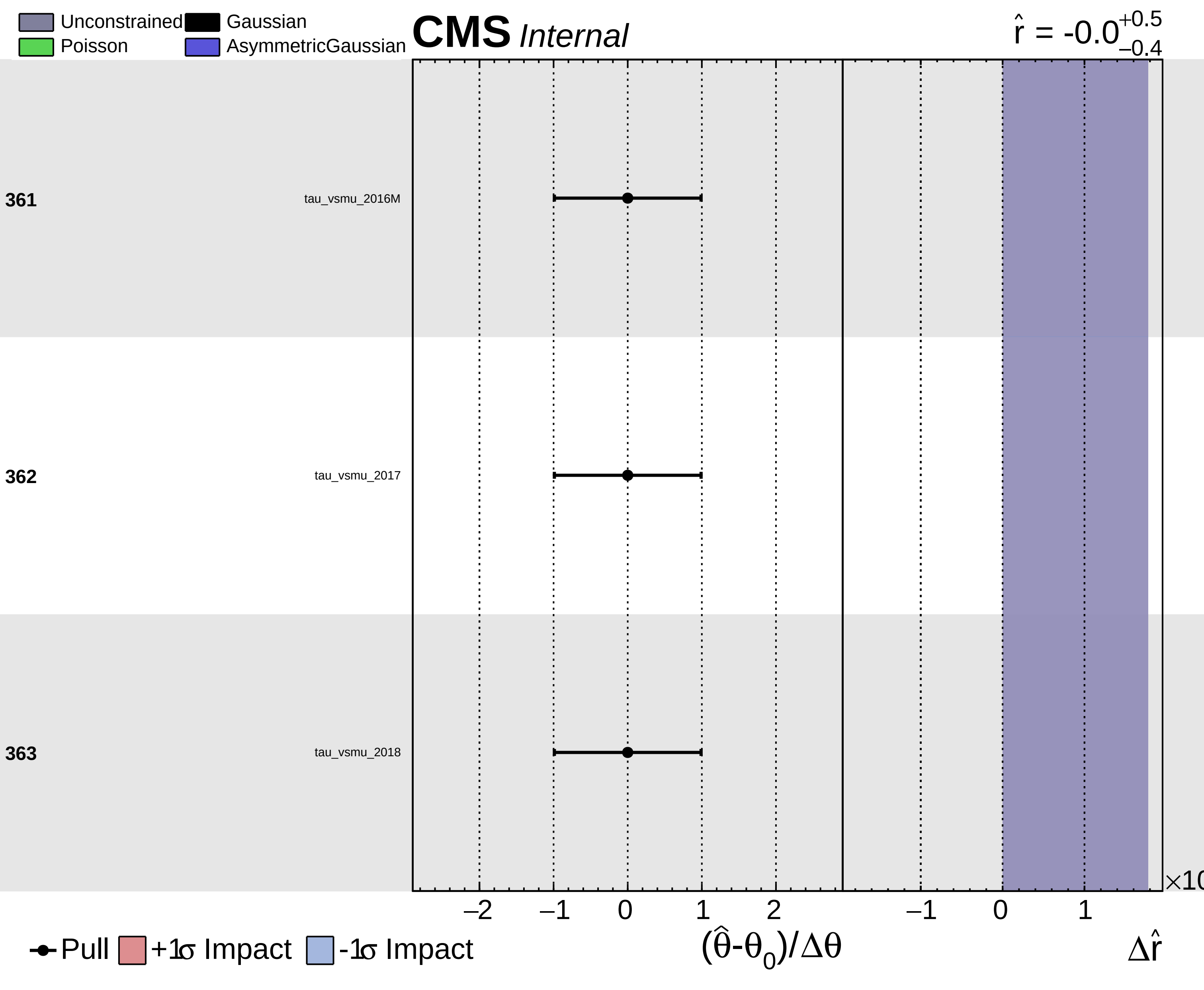
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
271 prop_binCRTT_electron2016M_bin9
272 prop_binCRWS_electron2016M_bin5
273 prop_binCRWS_electron2016M_bin8_DYJetsToLL_FxFx
274 prop_binCRWS_electron2016M_bin8_Fake
275 prop_binCRTT_muon2017_bin8
276 prop_binCRWS_electron2016M_bin7_Fake
277 prop_binCRTT_electron2017_bin3
278 prop_binCRTT_electron2016M_bin7
279 prop_binCRTT_electron2018_bin11
280 prop_binCRTT_electron2018_bin9
281 jer_2017
282 prop_binCRTT_electron2018_bin10
283 prop_binCRTT_electron2018_bin5
284 prop_binCRTT_muon2016M_bin5
285 prop_binSR_electron2016M_bin2_WZ
286 prop_binCRTT_muon2016M_bin9
287 prop_binCRWS_muon2017_bin9_WrongSign
288 prop_binCRTT_muon2018_bin11
289 prop_binSR_electron2016M_bin2_WrongSign
290 prop_binSR_electron2016M_bin3_VG
291 prop_binSR_electron2016M_bin2_TTTo2L2Nu
292 prop_binCRTT_muon2018_bin9
293 prop_binCRTT_muon2016M_bin2
294 prop_binSR_electron2016M_bin3_TTTo2L2Nu
295 lep_2016M
296 prop_binCRWS_muon2017_bin9_TTTo2L2Nu
297 tau_vsjet_2018
298 prop_binCRWS_electron2016M_bin8_WrongSign
299 prop_binCRWS_electron2016M_bin9_Fake
300 prop_binCRWS_muon2016M_bin8_TTTo2L2Nu
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.0 +0.5 −0.4
301 prop_binCRWS_muon2016M_bin8_WrongSign
302 prop_binSR_electron2016M_bin3_ZZtoLep
303 prop_binSR_electron2016M_bin2_VG
304 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
305 prop_binSR_electron2016M_bin2_ZZtoLep
306 lep_2017
307 prop_binSR_electron2016M_bin3_WrongSign
308 prop_binCRWS_electron2017_bin9_TTTo2L2Nu
309 prop_binSR_electron2016M_bin3_TVX
310 prop_binCRWS_muon2017_bin10_WrongSign
311 prop_binCRWS_muon2017_bin10_TTTo2L2Nu
312 prop_binCRWS_electron2016M_bin9_WrongSign
313 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
314 prop_binCRWS_electron2016M_bin7_WrongSign
315 prop_binSR_electron2016M_bin2_TVX
316 prop_binSR_electron2016M_bin3_Triboson
317 prop_binCRWS_electron2016M_bin7_TTTo2L2Nu
318 prop_binCRWS_muon2017_bin10_VG
319 prop_binSR_electron2016M_bin2_Triboson
320 prop_binCRWS_electron2016M_bin7_TVX
321 prop_binCRWS_electron2016M_bin8_VG
322 prop_binCRWS_muon2017_bin9_TVX
323 prop_binCRWS_electron2016M_bin9_Triboson
324 prop_binCRWS_electron2016M_bin8_TVX
325 prop_binCRWS_muon2016M_bin8_VG
326 prop_binCRWS_muon2016M_bin8_TVX
327 prop_binCRWS_electron2016M_bin8_ZZtoLep
328 prop_binCRWS_muon2017_bin9_Triboson
329 prop_binCRWS_electron2016M_bin8_Triboson
330 prop_binCRWS_muon2016M_bin8_Triboson
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.0 +0.5 −0.4
331 prop_binCRWS_muon2017_bin9_ZZtoLep
332 prop_binCRWS_muon2017_bin10_Triboson
333 prop_binCRWS_electron2016M_bin7_Triboson
334 prop_binCRWS_electron2016M_bin9_TVX
335 prop_binCRWS_electron2016M_bin9_ZZtoLep
336 prop_binCRWS_muon2017_bin10_TVX
337 prop_binCRWS_electron2017_bin9_ZZtoLep
338 prop_binCRWS_electron2016M_bin7_ZZtoLep
339 prop_binCRWS_muon2017_bin10_ZZtoLep
340 prop_binCRWS_electron2017_bin9_TVX
341 VBS_2016M
342 prop_binCRWS_electron2016M_bin8_WpWpJJ
343 FES
344 prop_binCRWS_muon2016M_bin8_WpWpJJ
345 prop_binCRWS_electron2017_bin9_Triboson
346 prop_binCRWS_muon2016M_bin8_ZZtoLep
347 VBS_2018
348 QCDScale_sig
349 prop_binCRWS_electron2016M_bin7_WpWpJJ
350 prop_binSR_electron2016M_bin3_WpWpJJ
351 prop_binCRWS_muon2017_bin10_WpWpJJ
352 prop_binCRWS_electron2016M_bin9_WpWpJJ
353 prop_binCRWS_electron2017_bin9_WpWpJJ
354 prop_binCRWS_muon2017_bin9_WpWpJJ
355 prop_binSR_electron2016M_bin2_WpWpJJ
356 VBS_2017
357 QCDScale_WZ
358 tau_vsele_2016M
359 tau_vsele_2017
360 tau_vsele_2018
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
361
tau_vsmu_2016M
362
tau_vsmu_2017
363
tau_vsmu_2018
–2 –1 0 1 2
–1 0 1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$